MIL-M-7507B(ASG)
5 FEBRUARY 1962
Superseding
MIL-M-7507A(ASG)
26 May 1953

MILITARY SPECIFICATION

MACHMETER, PITOT STATIC, RANGE .3 to 1.0, TYPE A-1

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 This specification covers one type of 2-3/4 inch round dial pitot static machmeter, designated type A-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED. STD. NO. 595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of

FSC 6610

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MS28105	Window, Dial - Aircraft Instrument Cover, Glass
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33638	Cases, Instrument, Flange-Mounted, Aircraft

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The machmeter furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Materials.-

3.2.1 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the machmeter, except where magnetic materials are essential.

3.2.2 Metals.- Metals shall be corrosion resistant, unless protected to resist corrosion during normal service life.

3.2.3 Protective treatment.- When materials are used in the construction of the machmeter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.4 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Standard MIL-STD-143, except as provided in the following paragraph.

3.2.4.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without

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alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design.- The machmeter shall be designed to indicate the ratio of the speed of the airplane to the speed of sound at any altitude and temperature.

3.4 Construction.- The machmeter shall be so constructed that no parts will work loose in service. The machmeter shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Pivots, bearings, and gears shall neither bind nor shake, and shall be as near frictionless as is practicable.

3.4.2 The machmeter shall be so constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases.

3.4.3 The design of the machmeter, luminous treatment, and method of adjusting the mechanism shall be subject to the approval of the procuring activity.

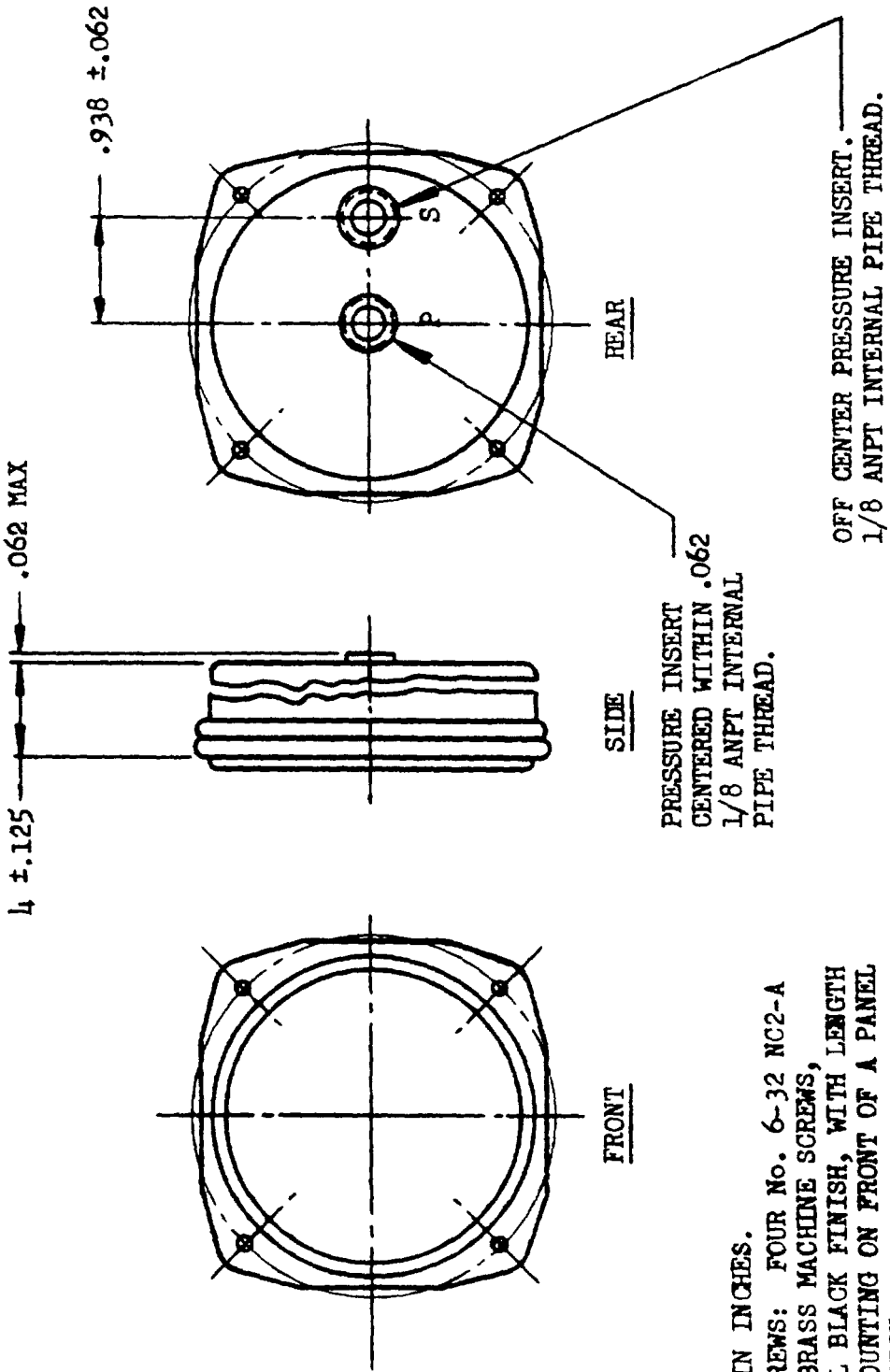
3.4.4 Case.- The machmeter case shall be in accordance with the 3-inch nominal size of Standard MS33638, except as shown on figure 1. The completely assembled case shall be airtight and raintight, except for the static pressure connection. The case shall be of suitable material, uniform in texture, and shall conform to the test requirements specified herein. (See 6.5.)

3.4.4.1 Bezel ring.- The bezel ring shall be held in place by means of screws properly secured by lock washers or similar devices acceptable to the procuring activity.

3.4.4.2 Plug.- The static and pitot pressure connections shall be provided with vented plugs of acceptable material and design. Each plug shall incorporate a vent consisting of a hole 1/16 ±1/64 inch in diameter.

3.4.4.3 Marking.- The back of the case, adjacent to the connections, shall be marked with the letter "P" for the pitot connection and the word "STATIC" printed twice, forming a ring around the static connection. (See 6.5.)

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DIMENSIONS IN INCHES.
MOUNTING SCREWS: FOUR No. 6-32 NC2-A ROUND HEAD BRASS MACHINE SCREWS, DURABLE DULL BLACK FINISH, WITH LENGTH TO PERMIT MOUNTING ON FRONT OF A PANEL .188 INCH THICK.

FIGURE 1. Case (see 6.5)

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3.4.4.3.1 The following shall be marked on the back of the case:

"CAUTION: DO NOT BLOW IN TUBES."

3.4.5 Cover glass.- The cover glass shall be in accordance with Part No. MS28105-16. The cover glass of the machmeter shall be properly sealed, and means shall be provided for the replacement of the cover glass by removal of the bezel.

3.4.6 Dial to cover glass distance.- The distance between the dial and the inside surface of the cover glass shall be the minimum practicable and shall not exceed 0.187 inch.

3.4.7 Dial.- The dial shall conform to figure 2. (See 6.5.)

3.4.7.1 Visibility of dial.- All graduations, numerals, pointer, and other markings on the dial shall be readable from any point within the frustum of a cone, the side of which makes an angle of not less than 30 degrees with the perpendicular to the dial, and the small diameter of which is the aperture of the dial.

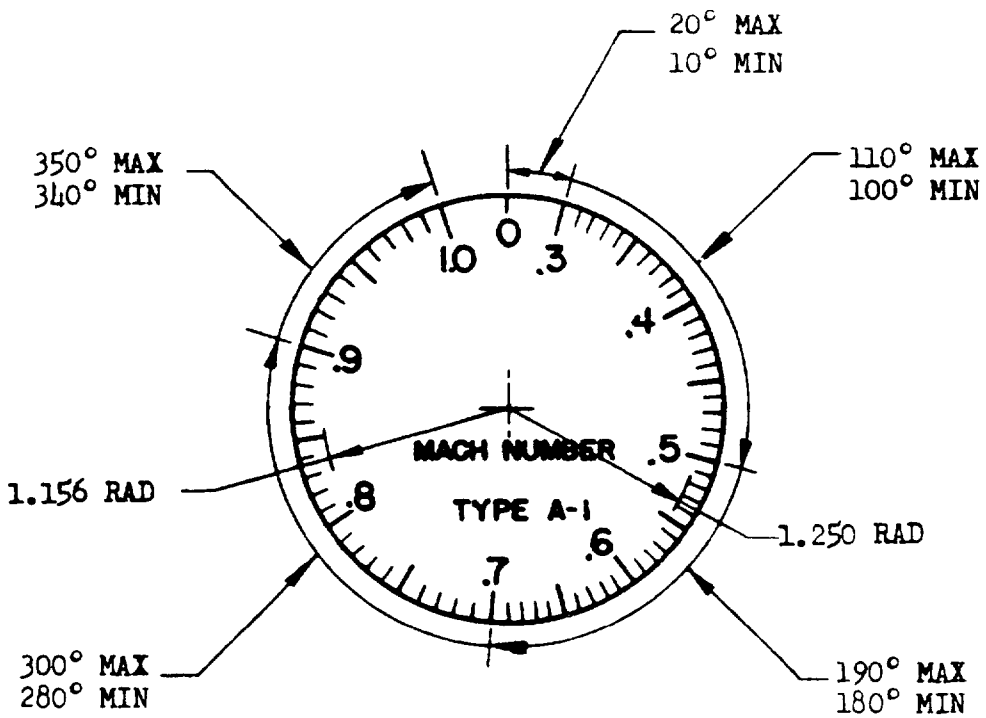
3.4.7.2 Dial markings.- All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters placed on the dial shall conform to Standard MS33558.

3.4.7.3 Numerals.- Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be approximately 1/64 inch.

3.4.7.4 Dial markings.- The dimensions of the markings shall be as follows:

	Height or length Inch ±0.010	Width of line or graduation Inch ±0.005
Numerals, except zero	0.312	0.031
Graduations corresponding to numerals (except zero graduation)	figure 2	0.031
Graduations corresponding to 0.350, 0.450, 0.550, 0.650, 0.750, 0.850, and 0.950	figure 2	0.031
Zero	0.156	0.016
Zero graduation	0.125	0.016
Minor graduations	figure 2	0.016
Lettering "MACH NUMBER"	0.125	0.016

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DIMENSIONS IN INCHES.

FIGURE 2. Dial (see 6.5)

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3.4.7.4.1 Air Force procurement.- For Air Force procurement, the above markings shall be finished in fluorescent-luminescent material in accordance with Specification MIL-L-25142.

3.4.7.4.2 Navy procurement.- For Navy procurement, the above markings shall be finished in lusterless white conforming to Color No. 37875 of Federal Standard No. 595.

3.4.7.5 Durable dull black.- The following markings shall be permanently and legibly marked on the dial. These markings, and all other markings not otherwise specified, shall be finished in durable dull black. The height of the markings shall be as follows:

	Height, inch
Lettering "TYPE A-1"	0.047

3.4.7.5.1 The background of the dial, unshaded part of the pointer, and the front of the bezel ring, shall be finished in durable dull black.

3.4.8 Pointer.- The pointer shall be in accordance with Part No. MS33585-3. For Air Force procurement, the shaded portion of the pointer shall be finished in fluorescent-luminescent material conforming to Specification MIL-L-25142. For Navy procurement, the shaded portion of the pointer shall be finished in lusterless white color No. 37875 of Federal Standard No. 595.

3.4.8.1 Limitation of pointer movement.- The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer shall not be permitted to rotate more than 350 degrees of arc clockwise from the zero graduation, when the machmeter is submitted to over-pressure tests. Suitable stops may also be incorporated in the machmeter mechanism to limit the counterclockwise rotation of the pointer to the zero graduation. The design of the stops shall be subject to the approval of the procuring activity. (See 6.5.)

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.6 Lubrication.- Where lubrication is necessary, it shall be accomplished by the use of lubricating oil which has been approved by the procuring activity. The lubricant shall be of such quality that satisfactory lubrication will be provided when the machmeter is operated at temperatures in the range of +71° to -54° C.

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3.7 Aluminum-alloy parts.- Aluminum-alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625, except that the dial, small holes, and case inserts, need not be anodized.

3.8 Weight.- The weight of the machmeter shall not exceed 1.5 pounds.

3.9 Dimensional tolerance.- Unless otherwise specified by the procuring activity, a tolerance of ±0.010 inch shall be permitted on all dimensions specified herein.

3.10 Screw threads.- Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter, shall conform to Specification MIL-S-7742.

3.11 Performance.- The machmeter shall be capable of passing the tests specified in section 4.

3.12 Application of pressure.-

3.12.1 The pitot pressure shall be applied to the inside of a diaphragm capsule and the static pressure to the interior of the case.

3.12.2 The machmeter shall be so designed that no errors will result because of varying humidity conditions.

3.12.3 The machmeter shall be so designed that errors in indication, due to changes in atmospheric temperature and the machmeter temperature, will be at a minimum.

3.12.4 The machmeter shall be so designed that no damage to the mechanism or change in performance will result from high speed such as exists during a power dive .

3.13 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.13.1 Use of MS or MIL designations.- MS or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.14 Mounting screws.- An envelope, containing the mounting screws shall be furnished with each machmeter (see figure 1). The following information shall be printed on tha face of the envelope.

IMPORTANT
This envelope contains mounting screws

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3.15 Workmanship.- The machmeter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of machmeters shall be classified as follows:

(a) Qualification tests (4.4)
(b) Acceptance tests (4.5)

4.3 Test conditions.-

4.3.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Vibration stand.- Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 500 and 3,000 cycles per minute (cpm) and shall subject the machmeter to vibration such that a point on the instrument case will describe, in a plane inclined 45 degrees to the horizontal plane, a circle of not less than 0.003 nor more than 0.005 inch in diameter, or a circle of not less than 0.018 nor more than 0.020 inch in diameter, as specified in the test.

4.3.3 Unless otherwise specified, the machmeter shall be tested in normal operating position and shall be lightly vibrated or tapped before a test reading is taken.

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4.4 Qualification testing.-

4.4.1 Prior qualification.- Unless otherwise specified by the procuring activity, machmeters which have not previously passed a qualification test, or which have passed the qualification tests and have been modified in any manner, shall satisfactorily pass a qualification test prior to the acceptance of any machmeter.

4.4.2 Sampling instructions.- The qualification test sample shall consist of one machmeter. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from the qualifying activity (see 6.3).

4.4.3 Qualification tests.- The qualification tests of the machmeter shall consist of all the tests specified under 4.6.

4.5 Acceptance tests.- Acceptance tests shall consist of the following:

(a)	Individual test	(4.5.1)
(b)	Sampling plans and tests	(4.5.2)

4.5.1 Individual tests.- Each machmeter shall be subjected to the following tests:

(a)	Examination of product	(4.6.1)
(b)	Leakage	(4.6.2)
(c)	Scale error at room temperature	(4.6.3)

4.5.2 Sampling plans and tests.-

4.5.2.1 Sampling plan A.- Five machmeters which have been manufactured under essentially the same conditions shall be selected at random from each quantity of 500, or fraction thereof, on the order or contract and shall be subjected to the following tests:

(a)	Individual tests	(4.5.1)
(b)	Magnetic property	(4.6.4)
(c)	Friction	(4.6.5)
(d)	Position error	(4.6.6)
(e)	Low temperature	(4.6.7)
(f)	High temperature	(4.6.8)
(g)	Vibration error	(4.6.9)
(h)	Over pressure	(4.6.10)
(i)	Seasoning	(4.6.11)
(j)	Pointer zero position error	(4.6.12)

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4.5.2.2 Sampling plan B.- Unless otherwise specified (see 6.2), 3 machmeters selected at random from the first 15 items on the contract or order shall be subjected to the following tests:

(a)	Sampling plan A tests	(4.5.2.1)
(b)	Vibration failure	(4.6.13)
(c)	Extreme low temperature exposure	(4.6.14)
(d)	High temperature exposure	(4.6.15)
(e)	Humidity	(4.6.16)
(f)	Fungus	(4.6.17)

4.6 Test methods.-

4.6.1 Examination of product.- The machmeter shall be inspected to determine compliance with this specification with respect to workmanship, finish, dimensions, and weight.

4.6.2 Leakage.-

4.6.2.1 Case leak.- The pitot and static pressure connections of the machmeter shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure. A suction of 15 inches Hg shall be slowly applied to the pitot and static pressure connections. With the source disconnected during a period of 10 seconds, the difference between the mercury levels in the manometer shall not change by more than 0.2 inch.

4.6.2.2 Diaphragm capsule leak.- The static connection shall be opened to the atmosphere, and a pressure sufficient to product approximately full scale deflection of the pointer, shall be applied to the pitot connection of the machmeter at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of 1 minute, the pointer shall not change its position by more than 0.02 MACH number.

4.6.3 Scale error at room temperature.- The machmeter shall be tested for scale errors at the points of the scale indicated in table I. During this test, the static pressure connection shall be suitably connected to an accurate barometer, a source of air pressure and suction, and to the suction side of water and mercury manometers. The pitot pressure connection shall be connected to a source of air pressure and suction, and to the pressure side of water and mercury manometers. The static pressure shall be adjusted to the standard value (29.92 inches Hg) corresponding to zero feet altitude, sea level standard, as specified in table I. The static pressure shall be held constant, and the scale error tests accomplished by subjecting the machmeter to the differential pressures specified to produce the readings, first with pressures increasing, then with pressures decreasing. With pressures increasing, the pressure shall be brought up to, but shall not exceed the pressures specified to obtain the

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desired reading; and with pressures decreasing, the pressure shall be brought down to, but shall not fall below, the pressure specified to obtain the desired reading. The errors at the test points shall not exceed the tolerances specified in table I. This procedure shall be repeated for each of the altitudes specified in table I.

TABLE I

Scale error tolerances

Pressure altitude Feet and inches Hg	Mach number	ΔP in. Hg	Tolerance Mach number
0	0.30	1.927	0.03
29.9212	0.40	3.489	0.03
	0.50	5.571	0.01
	0.60	8.243	0.01
	0.70	11.585	0.01
	0.80	15.691	0.01
	0.90	20.682	0.01
	1.00	26.717	0.01
5,000	0.30	1.603	0.03
24.8940	0.40	2.903	0.03
	0.50	4.635	0.01
	0.60	6.858	0.01
	0.70	9.639	0.01
	0.80	13.054	0.01
	0.90	17.207	0.01
	1.00	22.228	0.01
10,000	0.30	1.325	0.03
20.5736	0.40	2.399	0.03
	0.50	3.831	0.01
	0.60	5.668	0.01
	0.70	7.966	0.01
	0.80	10.789	0.01
	0.90	14.220	0.01
	1.00	18.370	0.01

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TABLE I (CONTINUED)

Scale error tolerances

Pressure altitude Feet and inches Hg	Mach number	Δ P in. Hg	Tolerance Mach number
15,000	0.30	1.087	0.03
16.8816	0.40	1.968	0.03
	0.50	3.143	0.01
	0.60	4.651	0.01
	0.70	6.537	0.01
	0.80	8.853	0.01
	0.90	11.669	0.01
	1.00	15.074	0.01
20,000	0.30	0.8852	0.03
13.7453	0.40	1.603	0.03
	0.50	2.559	0.01
	0.60	3.787	0.01
	0.70	5.322	0.01
	0.80	7.208	0.01
	0.90	9.501	0.01
	1.00	12.273	0.01
35,000	0.30	0.4531	0.03
7.0356	0.40	0.8204	0.03
	0.50	1.310	0.01
	0.60	1.938	0.01
	0.70	2.724	0.01
	0.80	3.689	0.01
	0.90	4.863	0.01
	1.00	6.282	0.01
50,000	0.30	0.2213	0.03
3.4370	0.40	0.4008	0.03
	0.50	0.6400	0.03
	0.60	0.9469	0.01
	0.70	1.331	0.01
	0.80	1.802	0.01
	0.90	2.376	0.01
	1.00	3.069	0.01

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4.6.4 Magnetic property.- The magnetic property of the machmeter shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long, in a magnetic field in air with a horizontal intensity of 0.18 ±0.01 oersted, when the machmeter is held in various positions on an east-west line 5 inches from the center of the magnet. (An aircraft compass with compensating magnets removed therefrom may be used as the free magnet for this test.) The maximum deflection of the magnet shall not exceed 1 degree.

4.6.5 Friction.- The machmeter shall be tested for friction at every other point in table I beginning with the first. The pressure shall be increased to bring the pointer approximately to the desired reading, and then held constant while two readings are taken, the first before the machmeter is tapped, and the second after the machmeter is tapped. The difference between any two such readings shall not exceed the tolerance specified in table II. The pointer shall move smoothly, while the pressure is varied uniformly, without vibration of the machmeter. The test may be combined with the test for scale errors.

4.6.6 Position error.- The machmeter reading shall be observed while the machmeter is held in any desired position, and tapped. This reading shall not differ from the reading when the machmeter is held in any other position, by more than the tolerance specified in table II. This test shall be accomplished at every other one of the points in table I beginning with the first.

4.6.6.1 Table I is calculated in accordance with the following:

$$\frac{\Delta P}{P} = \left(1 + \frac{M^2}{5}\right)^{\frac{7}{2}} - 1$$

ΔP = Differential pressure corresponding to given indicated airspeed.

P = Pressure corresponding to given altitude (same unit as ΔP).

M = Mach number

4.6.7 Low temperature.- The machmeter shall be tested for scale errors at each of the points in table I, beginning with the first. The test shall be as specified for room temperature, except that the machmeter shall be at a temperature of 54° ±2° C. The errors at the test points shall not exceed the tolerances specified in table I by more than the tolerance specified in table II.

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TABLE II

Allowable tolerances for sampling tests

Tests	Tolerance (Mach No.)
Friction:	
Tolerance	0.02
Position error:	
Tolerance	0.02
Vibration error:	
Altitude - 0 feet	
Tolerance at 0.30 Mach	0.01
Tolerance at 0.90 Mach	0.01
Altitude - 20,000 feet	
Tolerance at 0.30 Mach	0.01
Tolerance at 0.90 Mach	0.01
Vibration failure:	
Test altitude - 20,000 feet	
Tolerance at 0.60 Mach	0.02
Seasoning:	
Test altitude - 10,000 feet	
Tolerance	0.02
Low temperature:	
Tolerance	0.015
High temperature:	
Tolerance	0.015
Overpressure:	
Pitot pressure connection, 30 inches of mercury	0.02
Static pressure connection, 6 inches of water	
Scale error:	
Tolerance	0.02

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4.6.8 High temperature.- The machmeter shall be tested for scale errors as specified for room temperature, except that the machmeter shall be at a temperature of +71° ±2° C. The errors at the test points shall not exceed the tolerances specified in table I by more than the tolerance specified in table II.

4.6.9 Vibration error.- The machmeter shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in a horizontal plane, with the plane of the mounting lugs of the machmeter vertical. This test shall be conducted while the Mach numbers and altitudes specified in table II are applied to the machmeter. The machmeter shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm, in order to determine that the natural frequency of the machmeter does not occur in this frequency range. While the machmeter is being vibrated, the maximum amplitude of pointer vibration and the pointer variation, from its original position, shall not exceed the tolerance specified in table II.

4.6.10 Overpressure.- The pressure specified in table II shall be applied to the pitot connection of the machmeter for a period of 10 minutes. The pressure specified in table II shall be applied to the static connection of the machmeter for a period of 10 minutes. Not less than 5 minutes following the application of overpressures, the machmeter shall be tested for scale error as specified in 4.6.3. The errors at the test points shall not exceed the tolerances specified in table I by more than the tolerance specified in table II.

4.6.11 Seasoning.- The machmeter shall be subjected to 1,000 applications of a differential pressure sufficient to produce approximately full scale deflection. These applications shall be made at a rate not greater than 6 per minute. Not less than 1 hour following this test, the machmeter shall be tested for scale errors as specified in 4.6.3. The errors at the test points shall not exceed the tolerances specified in table I by more than the tolerance specified in table II.

4.6.12 Pointer zero position error.- The zero position of the pointer with no pressure applied to the indicator, shall be '0' within 1/16 inch.

4.6.13 Vibration failure.- The test shall be conducted while the machmeter is subjected to the altitude and Mach numbers specified in table I. The machmeter shall be subjected to vibration in accordance with procedure V of the Vibration test of Specification MIL-E-5272. At the completion of this test, the machmeter shall meet the requirements specified in 4.6.3. In addition, no looseness in the mechanism nor damage to any part of the machmeter shall result from the vibration.

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4.6.14 Extreme low-temperature exposure.- The machmeter shall be subjected to low temperature in accordance with procedure II of the Low temperature tests of Specification MIL-E-5272. Inspection of the machmeter is not required at the end of the first 48-hour period. At the conclusion of the exposure period and while still at this temperature, the machmeter shall be operated throughout its range without exhibiting sticking or erratic characteristics. The machmeter shall then be subjected to the low temperature test (4.6.7) while at the -55° ±2° C. There shall be no damage which would adversely affect subsequent operation.

4.6.15 High-temperature exposure.- The machmeter shall be subjected to high temperature in accordance with procedure II of the High temperature tests of Specification MIL-E-5272. The machmeter shall be maintained at +71° ±2° C for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the machmeter shall operate without indications of sticking or erratic action. After the chamber has been returned to room temperature and the machmeter has saturated at the room temperature for a minimum period of 4 hours, it shall pass the individual tests specified in 4.5.1. There shall be no damage which would adversely affect subsequent operation.

4.6.16 Humidity.- The machmeter shall be subjected to humidity in accordance with procedure I of the Humidity tests of Specification MIL-E-5272. At the conclusion of the 360-hour period, and after the machmeter has returned to room temperature, it shall be subjected to the test specified in paragraph 4.6.3. The scale errors shall not exceed the tolerances listed in table II for scale error at room temperature. The machmeter shall be examined and there shall be no evidence of corrosion or deterioration which will affect subsequent operation.

4.6.17 Fungus.- The machmeter shall be subjected to procedure I of the Fungus tests of Specification MIL-E-5272. The tests shall be conducted at the conclusion of qualification tests or Sampling plan B by disassembling one of the units to expose the internal mechanism to the fungus spores. At the end of the test period, the machmeter shall be carefully examined to ascertain that no fungus growth has occurred.

4.7 Packaging, packing, and marking.- Preparation for delivery shall be examined for conformance with section 5.

4.8 Rejection.- When Sampling tests are specified on a number of items that are selected as representative of a certain quantity and one or more of this number fails to meet the specified test, acceptance of all items shall be withheld until the extent and cause of failure is determined. The Individual tests need not be interrupted unless the defect is of such a nature that it will seriously affect the performance or safe use of the equipment, in which case all items shall be rejected. Rejected equipment shall not be resubmitted

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for inspection without furnishing all particulars concerning previous rejections and measures taken to overcome the defects. If investigation indicates the defect to be present in any item previously accepted, full particulars concerning the defect found, including recommendations for correction, shall be furnished to the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Machmeters shall be preserved in accordance with Method IA of Specification MIL-P-116 and packaged in accordance with Level A of Specification MIL-P-7936.

5.1.2 Level C.- Machmeters shall be preserved and packaged in accordance with manufacturers commercial practice.

5.2 Packing.- Machmeters shall be packed in accordance with Specification MIL-P-7936.

5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.3.1 Reinspection date marking.- Reinspection date marking shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use.- The type A-1 machmeter covered by this specification is intended to indicate the ratio of speed of the airplane to the speed of sound at the particular altitude and temperature condition.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Finish of dial and pointer (see 3.3.7.4 and 3.3.8).
(c) Dimensional tolerance (see 3.9).
(d) Whether Sampling plan B tests are required (see 4.5.2.2).
(e) Level of packaging and packing required (see 5.1 and 5.2).
(f) Reinspection date marking (see 5.3.1).

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the

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Qualified Products List is the Aeronautical Systems Division, Attention: ASNP, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions.-

6.4.1 "Scale errors" means the difference between the correct value of Mach number, corresponding to the differential pressure and static pressure, and the indication of the instrument.

6.4.2 "Amplitude" means the extent of vibratory movement as measured from one extreme position to the opposite extreme position.

6.5 International interest.- Certain provisions (see 3.4.4, 3.4.4.3, 3.4.7, 3.4.8.1, and figures 1 and 2) of this specification are the subject of international standardization agreement ABC Air Standard 10/9. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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International interest (see 6.5)

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